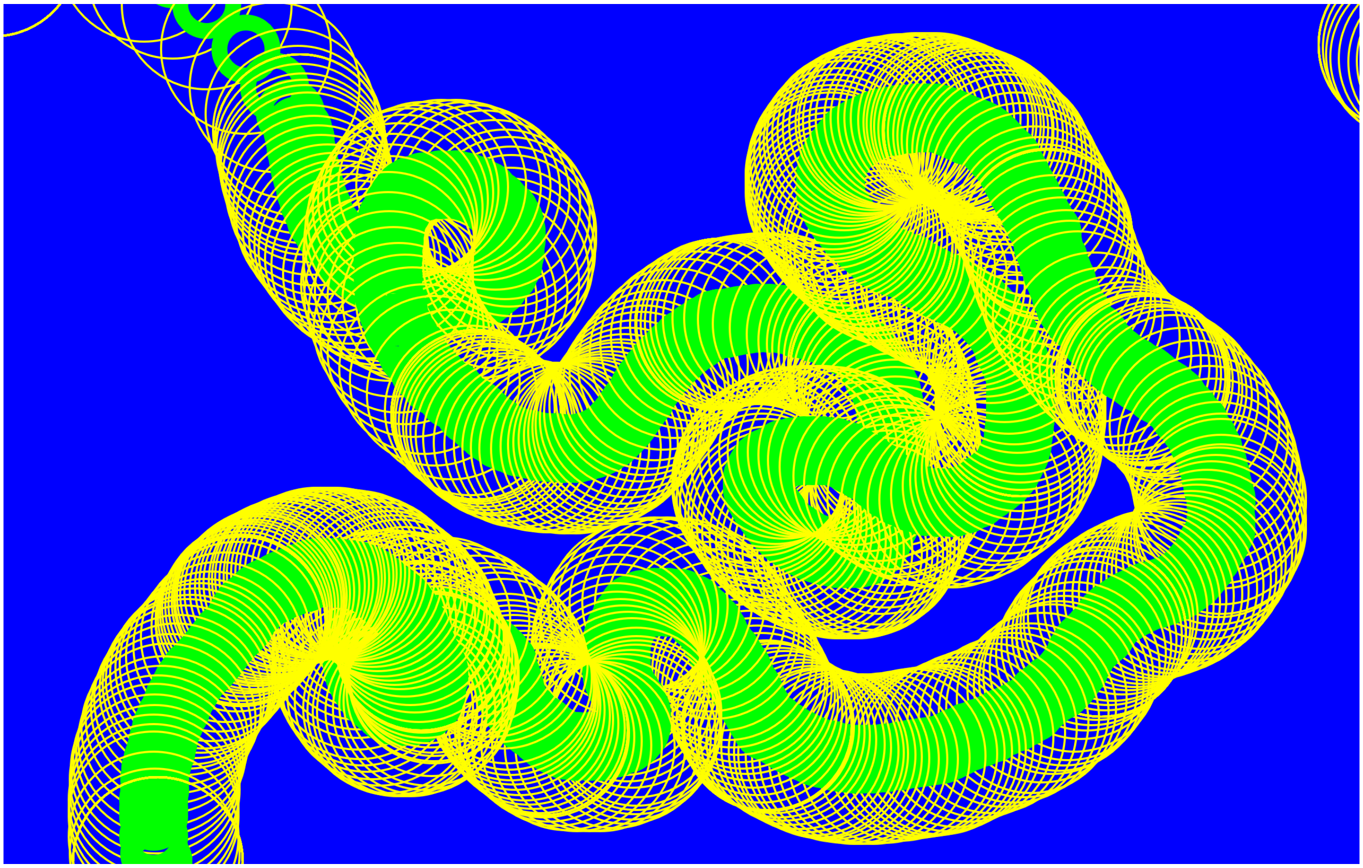
week-1

P5LIVE-Introduction Basics

1

basic forms - mouseX mouseY - frameRate

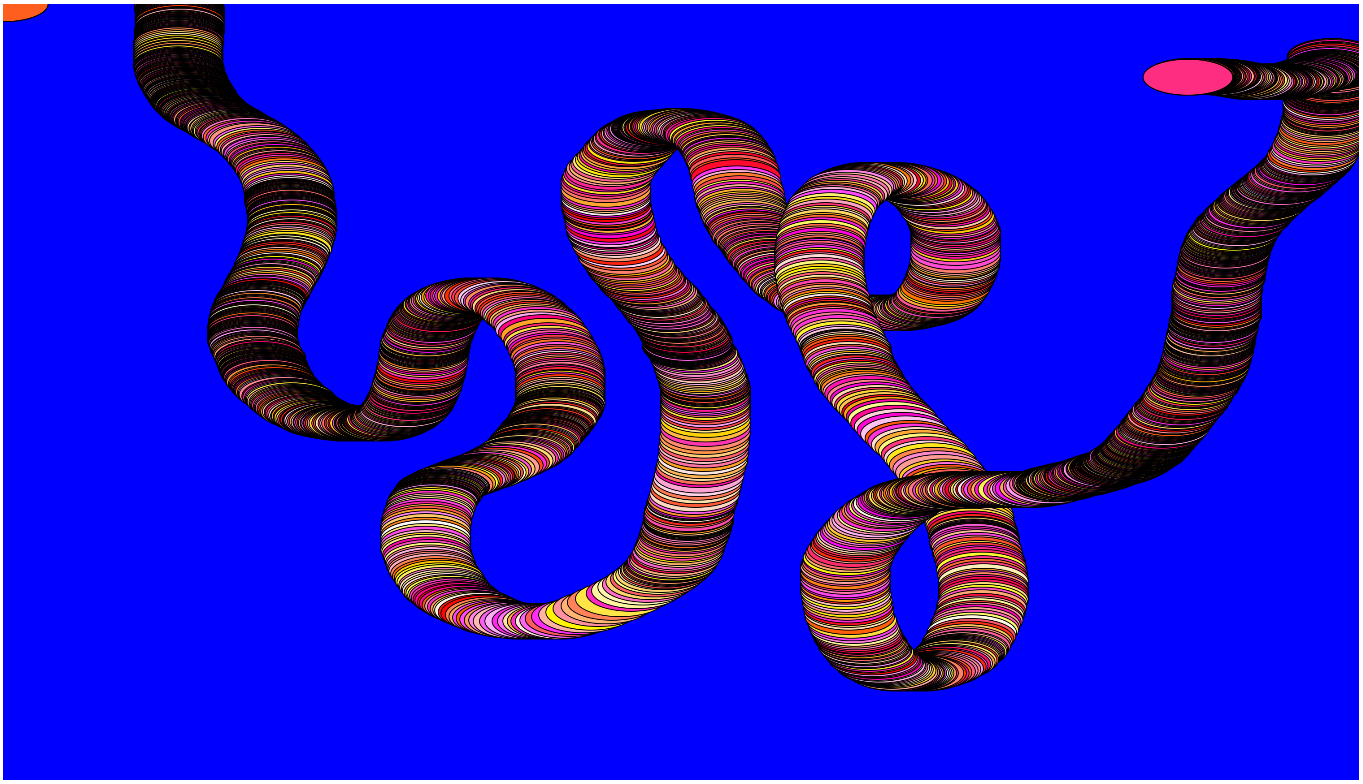


```
function setup () {
 createCanvas(windowWidth, windowHeight)
 background ("blue")
 rectMode (CENTER)
 frameRate (15)
}

function draw (){
noFill()
stroke ("yellow")
strokeWeight (2)
circle (mouseX, mouseY, 160, 5)
noFill()
stroke (0,255,0)
strokeWeight (15)
```

```
circle (mouseX, mouseY, 50)
}
```

2

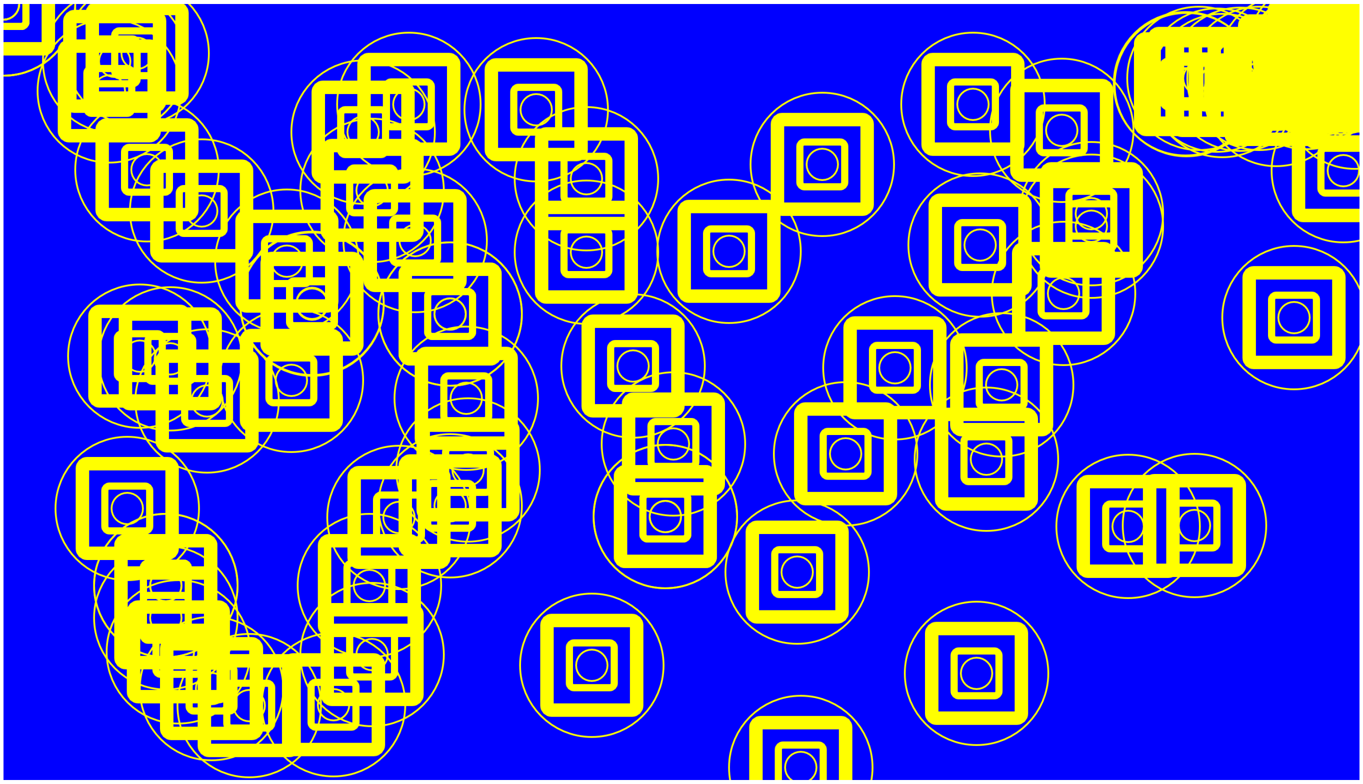
random()



```
function setup() {
     createCanvas(windowWidth, windowHeight)
    background("blue")

}

function draw() {
     //paint program very basic
     circlecolor = color(255, random(255),       random(255) )
     fill(circlecolor)
     ellipse(mouseX,mouseY,100,40)

}
```

3

basic forms

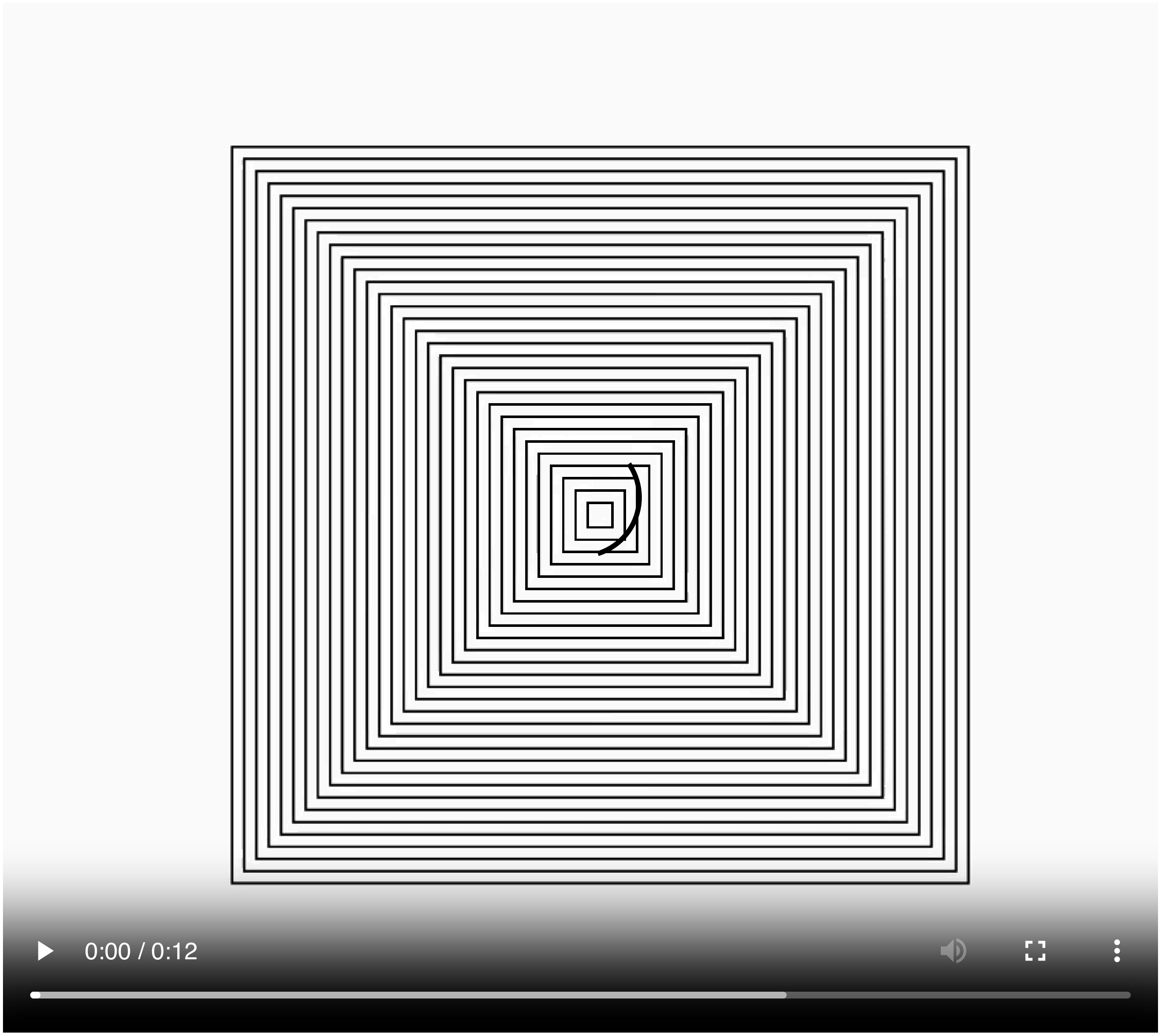


```
function setup() {
     createCanvas(windowWidth, windowHeight)
     background("blue")
     rectMode(CENTER)
     frameRate(5)

}

function draw() {
     noFill()
     stroke ("yellow")
     strokeWeight (2)
     circle(mouseX,mouseY,160,5)

     noFill()
     stroke ("yellow")
     strokeWeight (15)
     square(mouseX,mouseY,100,5)

     noFill()
     stroke ("yellow")
     strokeWeight (8)
     square(mouseX,mouseY,50,5)

     noFill()
```

```
    stroke ("yellow")
    strokeWeight (2)
    circle(mouseX,mouseY,35,5)
}
```

4

forLoop and variables

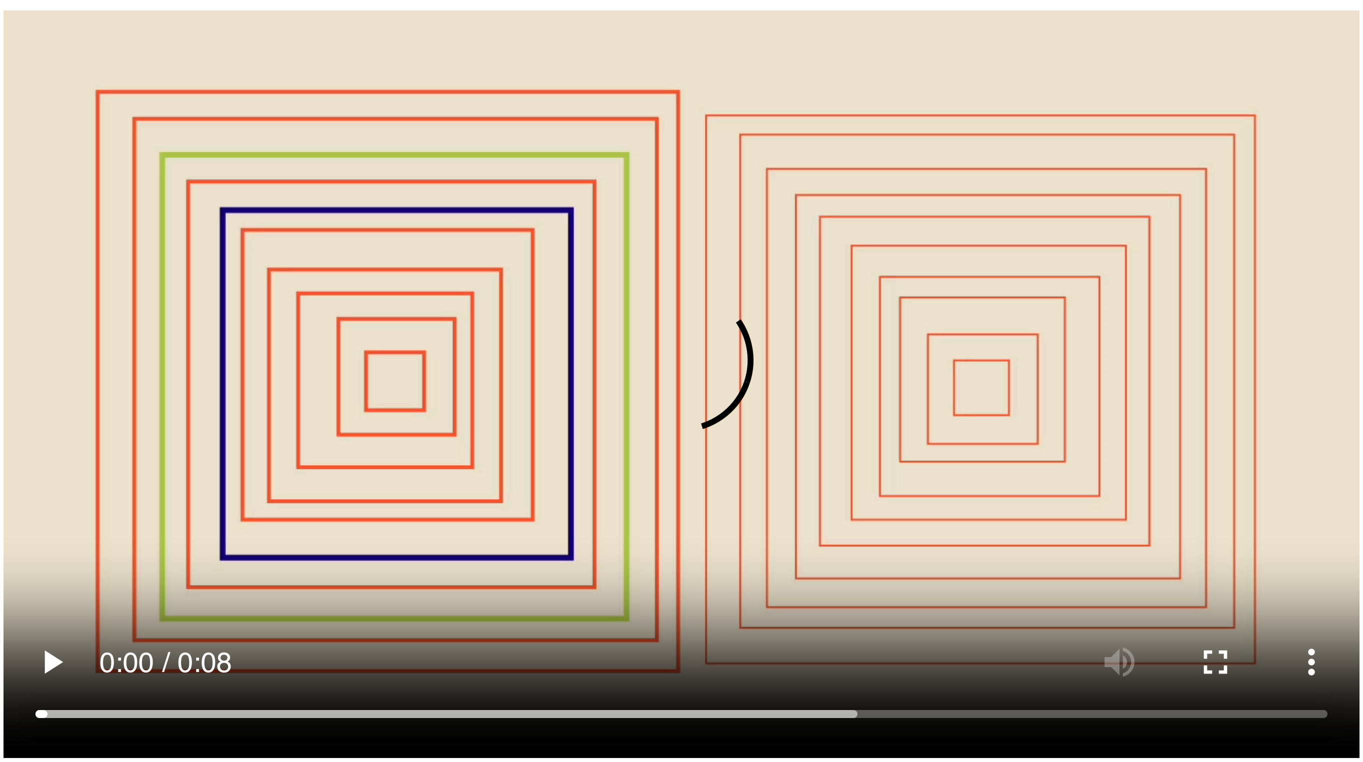



```
let dim = 300
let num = 30
let reduction = dim / num
let posX = 0
let posY = 0

function setup() {
    createCanvas(windowWidth, windowHeight)
```

```
    background(255)
    rectMode(CENTER)
    angleMode(DEGREES)
    frameRate(200)
    posX = width / 2
    posY = height / 2

}

function draw() {
    background(255)
    noFill()
    strokeWeight(1)
    stroke(0)
    dim = 200 + (sin(frameCount)*100)
    num = 30
    reduction = dim / num

    for(let i = 0; i < num; i++) {
        square(posX, posY, (dim) - (reduction * i))
    }


}
```

5

if-statement

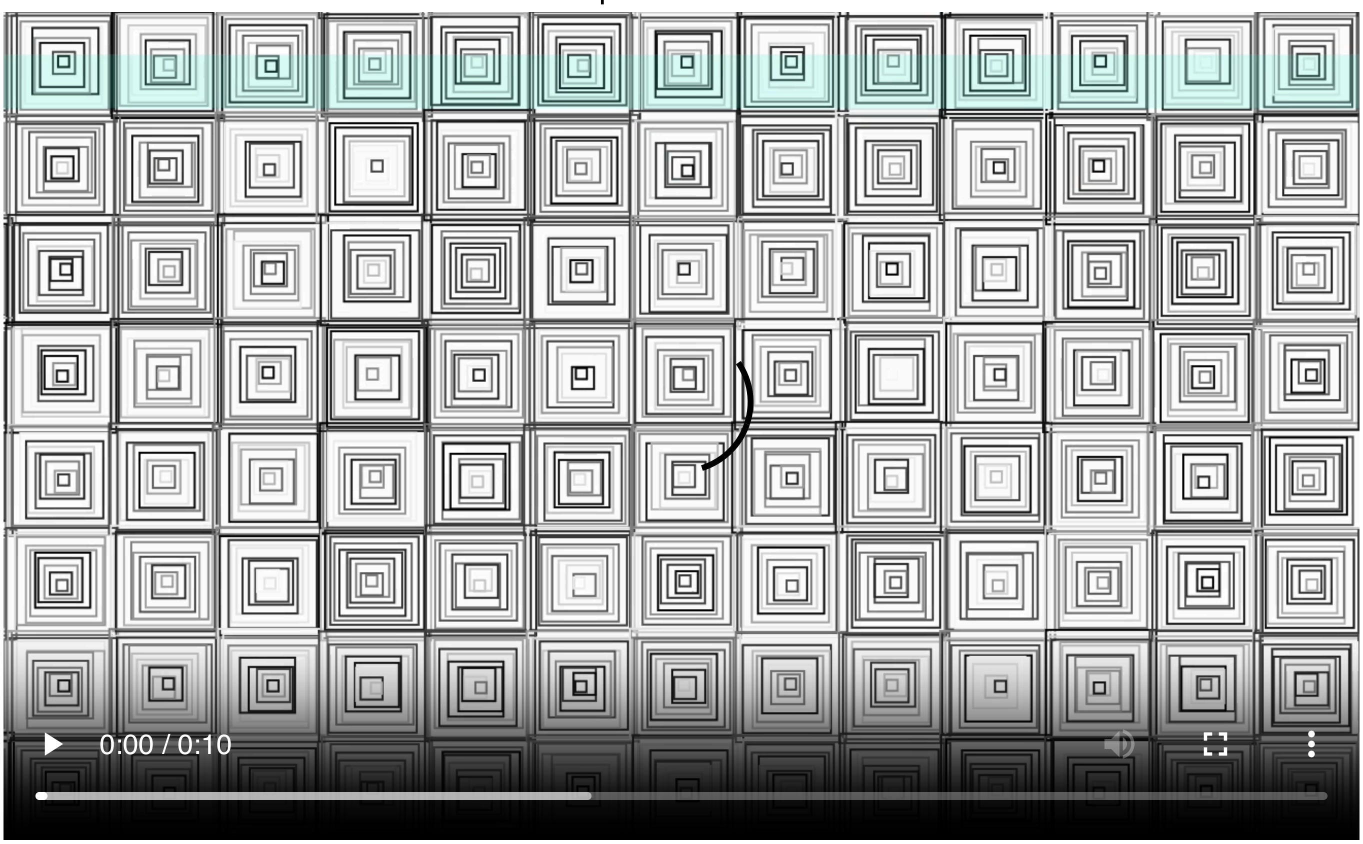


```
function setup() {
     createCanvas(windowWidth, windowHeight)
     background("blue")
     rectMode(CENTER)
     angleMode(DEGREES)
     //frameRate(2)
     posX = width / 2
     posY = height / 2
}

function draw() {
     background("#EFE7D4")
     noFill()
     strokeWeight(3)
     stroke(0)
     for(let i = 0; i < num; i++) {
          let offsetX = random(7)
          let offsetY = random(7)
          stroke("#FF5E32")
          strokeWeight(2)
          if(i == 2) {
               stroke("#B6CF4F")
               strokeWeight(3)
          }
          if(i == 4) {
               stroke("#1A0088")
               strokeWeight(3)
          }
          square(
               posX + offsetX,
               posY + offsetY,
               (dimX) - (reduction * i)
          )
     }

     dimX = 300 + sin(frameCount * 4) * 10
     num = 10
     reduction = dimX / num

     tmcs(1070, 435, 300, 10, 1)
}

//this function draws squares
//they move randomly a bit
function tmcs(x, y, dim, num, speed) {
```

```
    let dimension = dim + sin(frameCount * speed) * 10
    //num = 10
    let reduction = dimension / num

    for(let i = 0; i < num; i++) {
        let offsetX = random(7)
        let offsetY = random(7)
        stroke("#FF5E32")
        strokeWeight(1)
        square(
            x + offsetX,
            y + offsetY,
            (dimension) - (reduction * i)
        )
    }
}
```

6

create a function and call it within a for loop



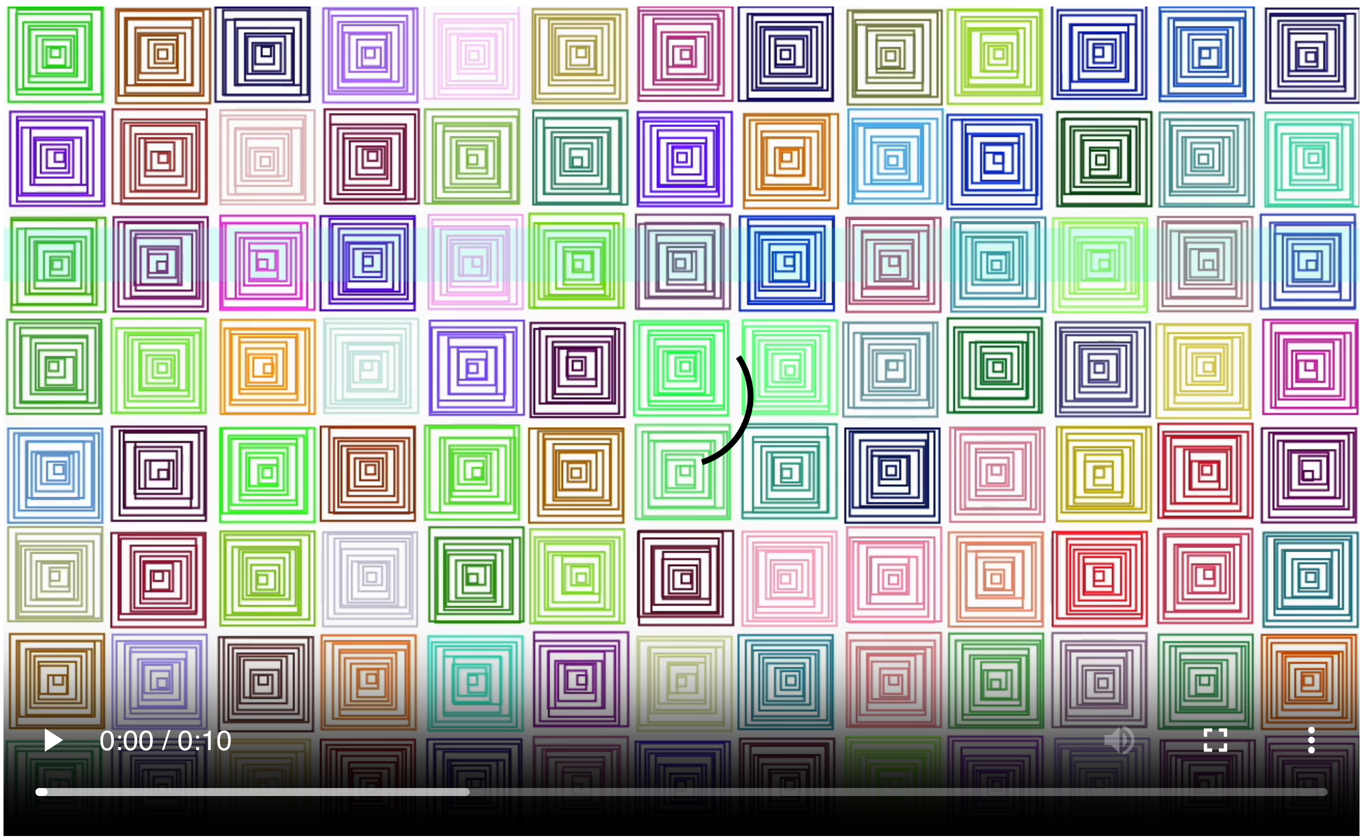

```
let dimX = 300
let dimY = dimX
let num = 10
let reduction = dimX / num
let posX = 0
```

```
let posY = 0

function setup() {
     createCanvas(windowWidth, windowHeight)
     background(0)
     rectMode(CENTER)
     angleMode(DEGREES)
     frameRate(5)
     posX = width / 2
     posY = height / 2
}

function draw() {
     background(255)
     noFill()
     strokeWeight(3)
     stroke(0)
     let numX = 30


     for(let i = 0; i < numX; i++) {
          let dimension = width / numX
          let posX = dimension / numX + (i * dimension)
          for(let j = 0; j < 30; j++){
               let posY = dimension / 2 + (j * dimension)
               tmcs(posX, posY, 2, dimension, 10)
          }

          //noLoop()

     }
}

//this function draws squares
//they move randomly a bit
function tmcs(x, y, speed, dim, num) {

     let dimension = dim + sin(frameCount * speed) * 10
     //num = 10
     let reduction = dimension / num

     for(let i = 0; i < num; i++) {
         let col = color(random(255))
          let offsetX = random(3)
          let offsetY = random(3)
          stroke(col)
```

```
            strokeWeight(1)
            square(
                x + offsetX,
                y + offsetY,
                (dimension) - (reduction * i),
            )
        }
    }
```

7

random color for each square

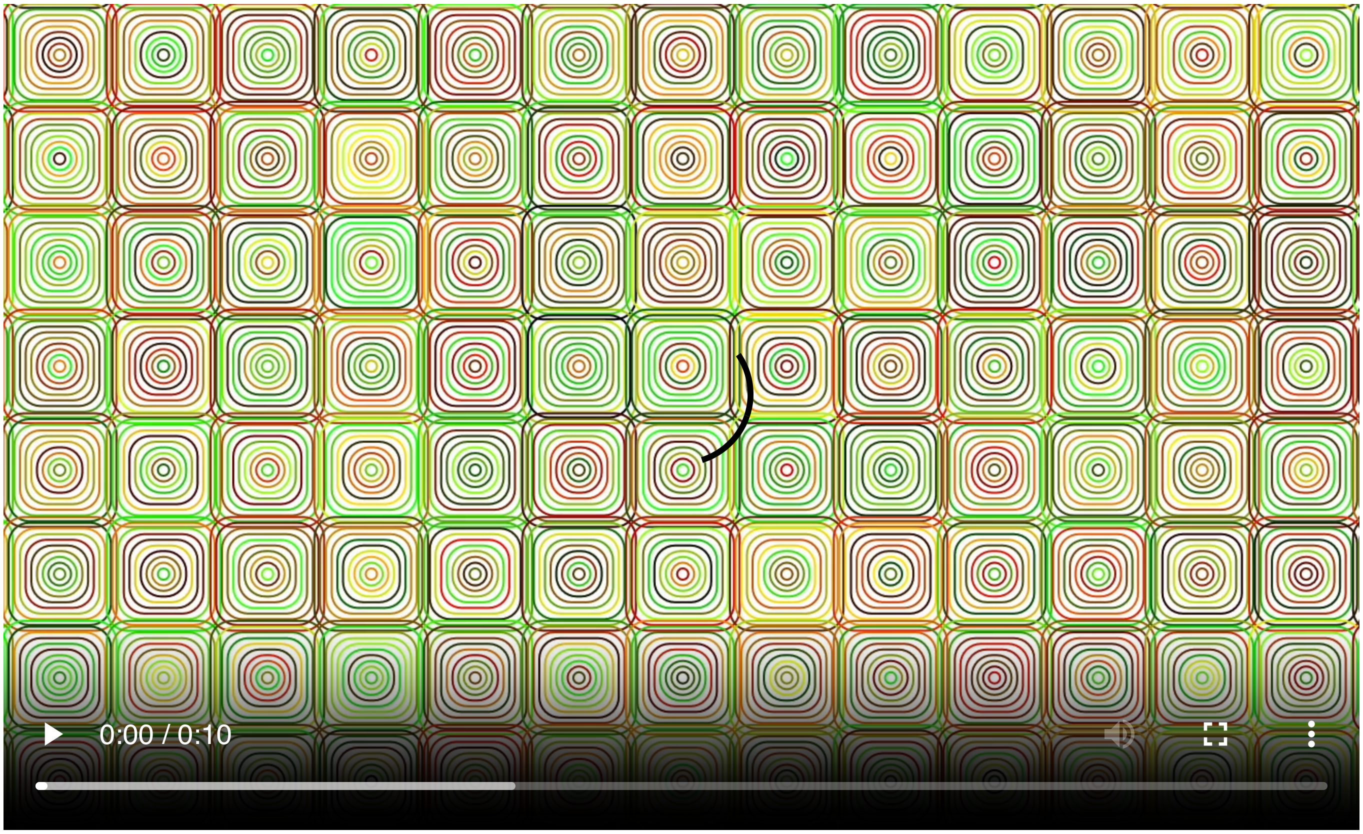



```
let dimX = 300
let dimY = dimX
let num = 10
let reduction = dimX / num
let posX = 0
let posY = 0

function setup() {
    createCanvas(windowWidth, windowHeight)
    background(0)
    rectMode(CENTER)
    angleMode(DEGREES)
    frameRate(5)
```

```
    posX = width / 2
    posY = height / 2
}

function draw() {
    background(255)
    noFill()
    strokeWeight(3)
    stroke(0)
    let numX = 30


    for(let i = 0; i < numX; i++) {
        let dimension = width / numX
        let posX = dimension / numX + (i * dimension)
        for(let j = 0; j < 30; j++){
            let posY = dimension / 2 + (j * dimension)
            tmcs(posX, posY, 50, dimension, 10)
        }

        //noLoop()

    }
}

//this function draws squares
//they move randomly a bit
function tmcs(x, y, speed, dim, num) {

    let dimension = dim + sin(frameCount * speed) * 10
    //num = 10
    let reduction = dimension / num
    let col = color(random(255),random(255),random(255))

    for(let i = 0; i < num; i++) {
        let offsetX = random(3)
        let offsetY = random(3)
        stroke(col)
        strokeWeight(1)
        square(
            x + offsetX,
            y + offsetY,
            (dimension) - (reduction * i),
        )
    }
}
```

8

random color within the for-loop - centered squares - rounded corners

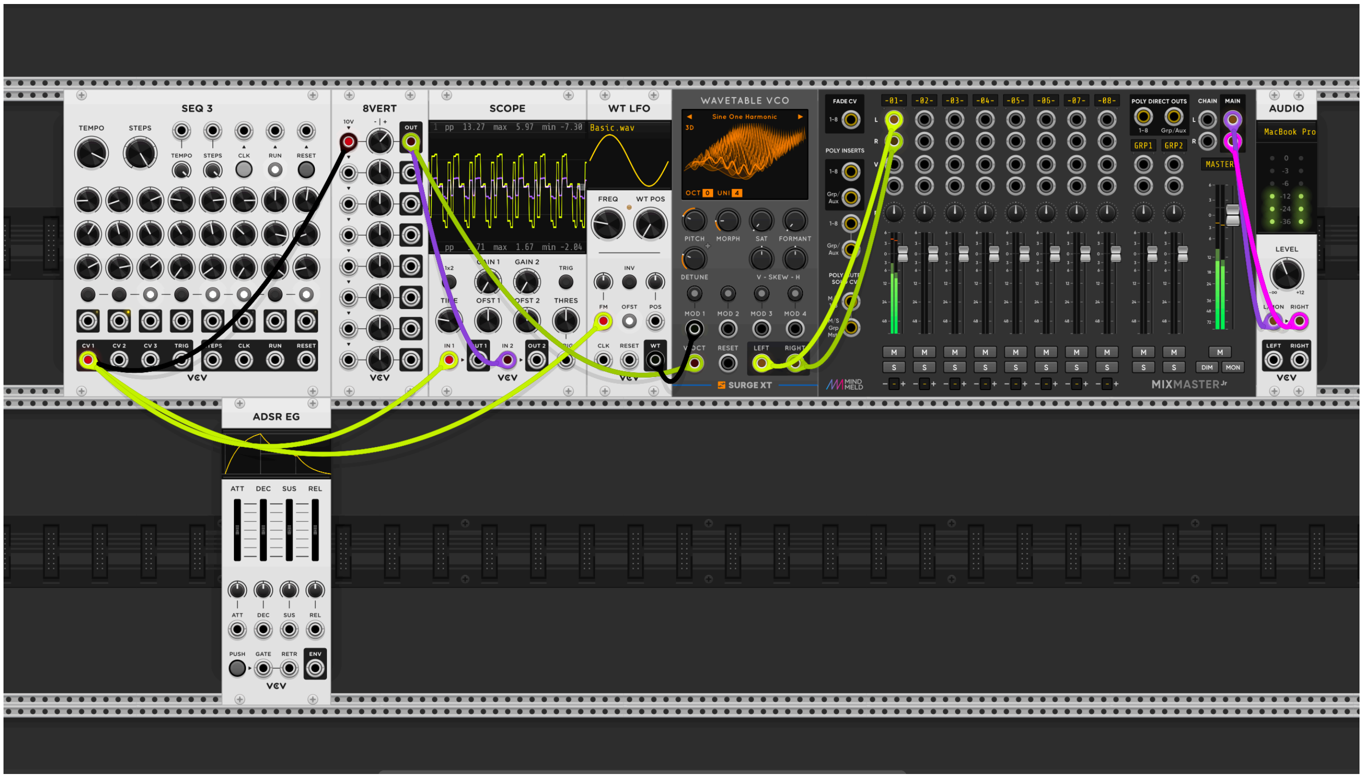



```
let dimX = 300
let dimY = dimX
let num = 10
let reduction = dimX / num
let posX = 0
let posY = 0

function setup() {
     createCanvas(windowWidth, windowHeight)
     background(0)
     rectMode(CENTER)
     angleMode(DEGREES)
     frameRate(5)
     posX = width / 2
     posY = height / 2
}

function draw() {
     background(255)
     noFill()
     strokeWeight(3)
     stroke(0)
```

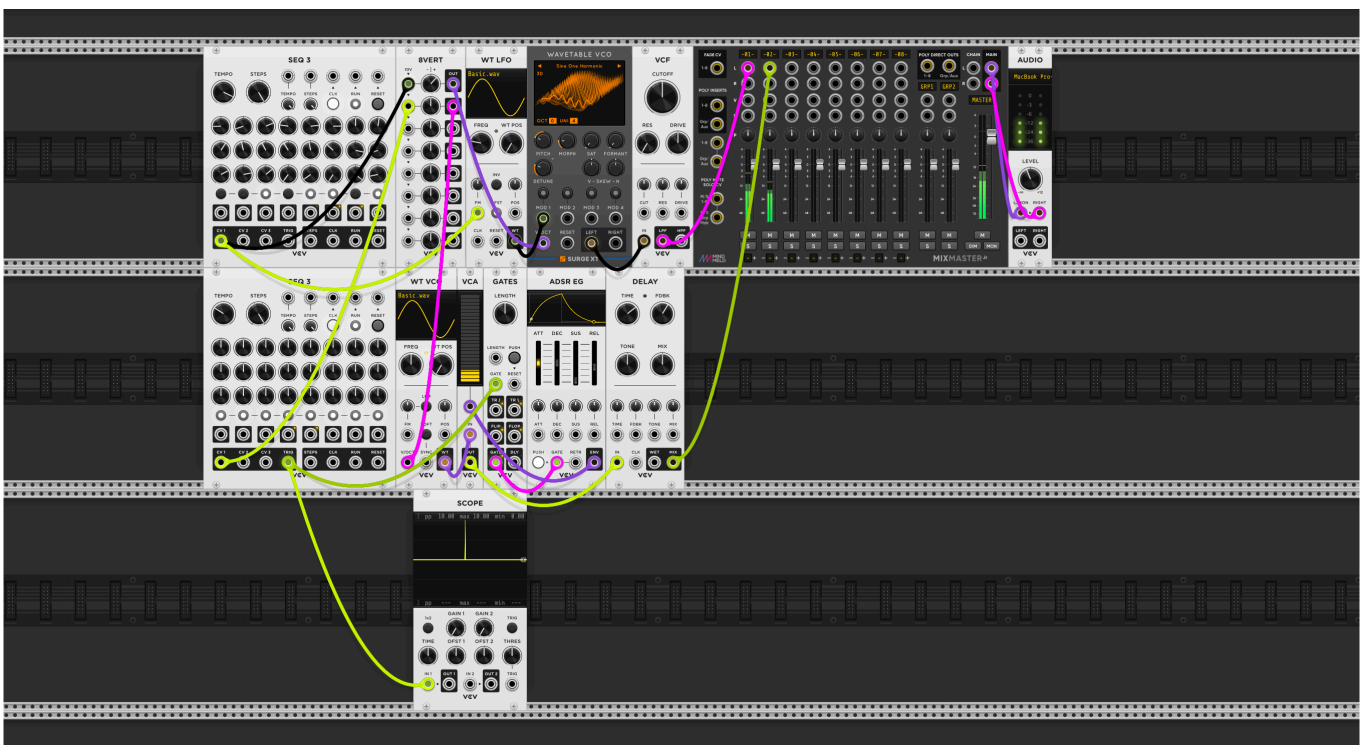
```
    let numX = 30


    for(let i = 0; i < numX; i++) {
        let dimension = width / numX
        let posX = dimension / numX + (i * dimension)
        for(let j = 0; j < 30; j++){
            let posY = dimension / 2 + (j * dimension)
            tmcs(posX, posY, 2, dimension, 10)
        }

        //noLoop()

    }
}

//this function draws squares
//they move randomly a bit
function tmcs(x, y, speed, dim, num) {

    let dimension = dim + sin(frameCount * speed) * 10
    //num = 10
    let reduction = dimension / num

    for(let i = 0; i < num; i++) {
       let col = color(random(255),random(255),0)
        let offsetX = random(3)
        let offsetY = random(3)
        stroke(col)
        strokeWeight(1)
        square(
            x /*+ offsetX*/,
            y /*+ offsetY*/,
            (dimension) - (reduction * i),10
        )
    }
}
```

VCV-Rack-Introduction Basics

1


SEQ 3
8VERT
SCOPE
WT LFO
WAVETABLE VCO
Sine One Harmonic
SURGE XT
MIXMASTER
AUDIO
MacBook Pro
ADSR EG
Basic.wav

2


SEQ 3
8VERT
WT LFO
WAVETABLE VCO
VCF
AUDIO
WT VCO
VCA
GATES
ADSR EG
DELAY
SCOPE